SQL- Project-4 (Case Study on Workforce Data Analysis)

Task-1:

1. Investigating the job market based on company size in 2021:
   - SELECT company_size, COUNT(*) AS total_employees FROM workforce WHERE work_year = 2021 GROUP BY company_size;
   - Output:

	company_size	total_employees
1	NULL	1
2	L	1
3	M	1

2. Top 3 job titles with the highest average salary for part-time positions in 2023:
   - SELECT TOP 3 job_title, AVG(salary_in_usd) AS avg_salary FROM workforce WHERE employment_type = 'PT' AND work_year = 2023 GROUP BY job_title HAVING COUNT(*) > 50 ORDER BY avg_salary DESC;
   - Output:

job_title	avg_salary

3. Countries where mid -level salary is higher than the overall mid -level salary in 2023:
   - SELECT company_location AS country, AVG(salary_in_usd) AS avg_mid_salary FROM workforce WHERE experience_level = 'MI' AND work_year = 2023 GROUP BY company_location HAVING AVG(salary_in_usd) >(SELECT AVG(salary_in_usd)FROM workforce WHERE experience_level = 'MI' AND work_year = 2023);
   - Output:

	country	avg_mid_salary
1	IN	70000

4. Highest and lowest average salary locations for senior-level employees in 2023:
   - Highest average salary:
   - SELECT TOP 1 company_location, AVG(salary_in_usd) AS avg_salary FROM workforce WHERE experience_level = 'SE' AND work_year = 2023 GROUP BY company_location ORDER BY avg_salary DESC;
   - Output:

	company_location	avg_salary
1	US	150000

   - Lowest average salary:
   - SELECT TOP 1company_location, AVG(salary_in_usd) AS avg_salary FROM workforce WHERE experience_level = 'SE' AND work_year = 2023 GROUP BY company_location ORDER BY avg_salary ASC;
   - Output:

	company_location	avg_salary
1	US	150000

5. Salary growth rates by job title:
   - SELECT

```
  job_title,
  AVG(CASE WHEN work_year = 2023 THEN salary_in_usd END) AS
avg_2023_salary,
  AVG(CASE WHEN work_year = 2024 THEN salary_in_usd END) AS
avg_2024_salary,
  (
    (AVG(CASE WHEN work_year = 2024 THEN salary_in_usd END)
    -AVG(CASE WHEN work_year = 2023 THEN salary_in_usd END)
    )/AVG(CASE WHEN work_year = 2023 THEN salary_in_usd END)
  ) * 100 AS growth_percentage
FROM workforce
WHERE work_year IN (2023, 2024)
```

GROUP BY job_title;

- Output:

	job_title	avg_2023_salary	avg_2024_salary	growth_percentage
1	Data Analyst	75000	NULL	NULL
2	Data Scientist	150000	NULL	NULL
3	Engineering Manager	180000	NULL	NULL
4	Part-Time Data Analyst	20250	NULL	NULL
5	QA Tester	NULL	55200	NULL
6	Software Engineer	55000	NULL	NULL

6. Top three countries with the highest salary growth for entrylevel roles from 2020 to 2023:

- SELECT TOP 3

```
company_location AS country,

AVG(CASE WHEN work_year = 2020 THEN salary_in_usd END) AS
avg_2020_salary,
AVG(CASE WHEN work_year = 2023 THEN salary_in_usd END) AS
avg_2023_salary,

(
    (AVG(CASE WHEN work_year = 2023 THEN salary_in_usd END) -
     AVG(CASE WHEN work_year = 2020 THEN salary_in_usd END))
    /AVG(CASE WHEN work_year = 2020 THEN salary_in_usd END)
) * 100 AS growth_percentage

FROM workforce
WHERE experience_level = 'EN'
  AND work_year IN (2020, 2023)
GROUP BY company_location
HAVING COUNT(*) > 50
ORDER BY growth_percentage DESC;
```

- Output:

country	avg_2020_salary	avg_2023_salary	growth_percentage

7. Updating remote work ratio for employees earning more than $90,000 in the US and AU:
   - UPDATE workforce SET remote_ratio = 100 WHERE salary_in_usd > 90000 AND employee_residence IN ('US', 'AU');

8. Salary updates based on percentage increases by level in 2024:
   - UPDATE workforce

```
SET salary_in_usd = salary_in_usd *
  CASE
    WHEN experience_level = 'SE' THEN 1.22   -- +22%
    WHEN experience_level = 'MI' THEN 1.30   -- +30%
    WHEN experience_level = 'EN' THEN 1.15   -- +15%
    WHEN experience_level = 'EX' THEN 1.25   -- +25%
    ELSE 1
  END
WHERE work_year = 2024;
```

9. Year with the highest average salary for each job title:
   - SELECT job_title, work_year, avg_salary

```
FROM (
  SELECT
    job_title,
    work_year,
    AVG(salary_in_usd) AS avg_salary,
    ROW_NUMBER() OVER (PARTITION BY job_title ORDER BY
AVG(salary_in_usd) DESC) AS rn
  FROM workforce
  GROUP BY job_title, work_year
) AS t
WHERE rn = 1;
```

- Output:

	job_title	work_year	avg_salary
1	Business Analyst	2021	40000
2	Data Analyst	2023	75000
3	Data Scientist	2023	150000
4	DevOps Engineer	2021	125000
5	Engineering Manager	2023	180000
6	Junior Developer	2020	35000
7	ML Engineer	2022	130000
8	Part-Time Data Analyst	2023	20250
9	Product Manager	2022	90000
10	QA Tester	2024	63480
11	Software Engineer	2023	55000

10. Percentage of employment types for different job titles:

- SELECT

```
    job_title,
   (SUM(CASE WHEN employment_type = 'FT' THEN 1 ELSE 0 END) * 100.0 /
    COUNT(*)) AS full_time_percentage,

   (SUM(CASE WHEN employment_type = 'PT' THEN 1 ELSE 0 END) * 100.0 /
    COUNT(*)) AS part_time_percentage,

   (SUM(CASE WHEN employment_type = 'CT' THEN 1 ELSE 0 END) * 100.0 /
    COUNT(*)) AS contract_percentage,

   (SUM(CASE WHEN employment_type = 'FL' THEN 1 ELSE 0 END) * 100.0 /
    COUNT(*)) AS freelance_percentage

FROM workforce
GROUP BY job_title;
```

- Output:

	job_title	full_time_percentage	part_time_percentage	contract_percentage	freelance_percentage
1	Business Analyst	0.000000000000	100.000000000000	0.000000000000	0.000000000000
2	Data Analyst	100.000000000000	0.000000000000	0.000000000000	0.000000000000
3	Data Scientist	100.000000000000	0.000000000000	0.000000000000	0.000000000000
4	DevOps Engineer	100.000000000000	0.000000000000	0.000000000000	0.000000000000
5	Engineering Manager	100.000000000000	0.000000000000	0.000000000000	0.000000000000
6	Junior Developer	100.000000000000	0.000000000000	0.000000000000	0.000000000000
7	ML Engineer	100.000000000000	0.000000000000	0.000000000000	0.000000000000
8	Part-Time Data Analyst	0.000000000000	100.000000000000	0.000000000000	0.000000000000
9	Product Manager	100.000000000000	0.000000000000	0.000000000000	0.000000000000
10	QA Tester	100.000000000000	0.000000000000	0.000000000000	0.000000000000
11	Software Engineer	100.000000000000	0.000000000000	0.000000000000	0.000000000000

11. COUNTRIES OFFERING FULL REMOTE WORK FOR MANAGERS WITH SALARIES OVER $90,000:

- SELECT DISTINCT employee_residence AS country
  FROM workforce
  WHERE job_title LIKE '%Manager%'
  AND salary_in_usd > 90000
  AND remote_ratio = 100;
- Output:

	country
1	US

12. Top 5 countries with the most large companies:

- SELECT TOP 5
  company_location,
  COUNT(*) AS large_company_count
  FROM workforce
  WHERE company_size = 'L'
  GROUP BY company_location
  ORDER BY large_company_count DESC;
- Output:

	company_location	large_company_count
1	CA	2
2	US	2
3	NL	1
4	IN	1
5	DE	1

13. Percentage of employees with fully remote roles earning more than $100,000:

- SELECT (
  COUNT(CASE WHEN remote_ratio = 100 AND salary_in_usd > 100000
  THEN 1 END)
  * 100.0 / COUNT(*)
) AS percentage_fully_remote_high_earners FROM workforce;
- Output:

	percentage_fully_remote_high_earners
1	17.647058823529

14. Locations where entry-level average salaries exceed market average for entry level:

- SELECT
  company_location,
  AVG(salary_in_usd) AS avg_entry_salary
  FROM workforce
  WHERE experience_level = 'EN'
  GROUP BY company_location
  HAVING AVG(salary_in_usd) >
  (
  SELECT AVG(salary_in_usd)
  FROM workforce
  WHERE experience_level = 'EN'
  );
- Output:

	company_location	avg_entry_salary
1	GB	55000

15. Countries paying the maximum average salary for each job title:

- SELECT job_title, employee_residence AS country, avg_salary

```
FROM (
  SELECT
    job_title,
    employee_residence,
    AVG(salary_in_usd) AS avg_salary,
    ROW_NUMBER() OVER (
      PARTITION BY job_title
      ORDER BY AVG(salary_in_usd) DESC
    ) AS rn
  FROM workforce
  GROUP BY job_title, employee_residence
) AS t
WHERE rn = 1;
```

- Output:

	job_title	country	avg_salary
1	Business Analyst	AU	40000
2	Data Analyst	IN	70000
3	Data Scientist	US	150000
4	DevOps Engineer	NL	125000
5	Engineering Manager	US	180000
6	Junior Developer	IN	35000
7	ML Engineer	CA	130000
8	Part-Time Data Analyst	CA	21000
9	Product Manager	DE	90000
10	QA Tester	PH	63480
11	Software Engineer	GB	55000

16. Countries with sustained salary growth over three years:

- SELECT company_location
  FROM workforce
  WHERE work_year IN (2021, 2022, 2023)
  GROUP BY company_location
  HAVING
  AVG(CASE WHEN work_year = 2021 THEN salary_in_usd END) <
  AVG(CASE WHEN work_year = 2022 THEN salary_in_usd END)
  AND
  AVG(CASE WHEN work_year = 2022 THEN salary_in_usd END) <
  AVG(CASE WHEN work_year = 2023 THEN salary_in_usd END);
- Output:

	company_location
1	US

17. PERCENTAGE OF FULLY REMOTE WORK BY EXPERIENCE LEVEL (2021 VS 2024):

- SELECT
  experience_level,

  (SUM(CASE WHEN work_year = 2021 AND remote_ratio = 100 THEN 1 END)
  * 100.0 /
  SUM(CASE WHEN work_year = 2021 THEN 1 END)) AS pct_2021,

  (SUM(CASE WHEN work_year = 2024 AND remote_ratio = 100 THEN 1 END)
  * 100.0 /
  SUM(CASE WHEN work_year = 2024 THEN 1 END)) AS pct_2024
  FROM workforce
  WHERE work_year IN (2021, 2024)
  GROUP BY experience_level;
- Output:

	experience_level	pct_2021	pct_2024
1	EN	NULL	NULL
2	MI	NULL	NULL
3	SE	100.000000000000	NULL

18. Average salary increase percentage by experience level and job title (2023 to 2024):

- SELECT

```
    experience_level,
    job_title,

    AVG(CASE WHEN work_year = 2023 THEN salary_in_usd END) AS avg_2023_salary,
    AVG(CASE WHEN work_year = 2024 THEN salary_in_usd END) AS avg_2024_salary,

    (
      (AVG(CASE WHEN work_year = 2024 THEN salary_in_usd END) -
       AVG(CASE WHEN work_year = 2023 THEN salary_in_usd END))
      /
      AVG(CASE WHEN work_year = 2023 THEN salary_in_usd END)
    ) * 100 AS salary_increase_percentage
FROM workforce
WHERE work_year IN (2023, 2024)
GROUP BY experience_level, job_title;
```

- Output:

	experience_level	job_title	avg_2023_salary	avg_2024_salary	salary_increase_percentage
1	MI	Data Analyst	75000	NULL	NULL
2	SE	Data Scientist	150000	NULL	NULL
3	EX	Engineering Manager	180000	NULL	NULL
4	EN	Part-Time Data Analyst	18000	NULL	NULL
5	MI	Part-Time Data Analyst	21000	NULL	NULL
6	EN	QA Tester	NULL	63480	NULL
7	EN	Software Engineer	55000	NULL	NULL

19. Role-based access control for employees based on experience level:

- DECLARE @current_user_experience_level VARCHAR(5);

```
-- Example (you can change this value based on login):
SET @current_user_experience_level = 'MI';
-- Options: 'EN', 'MI', 'SE', 'EX'
-- Step 2: RBAC — Return only allowed records
SELECT *
FROM workforce
WHERE experience_level = @current_user_experience_level;
```

- Output:

	work_year	experience_level	employment_type	job_title	salary	salary_currency	salary_in_usd	employee_residence	remote_ratio	company_location	company_size
1	2023	MI	FT	Data Analyst	70000	USD	70000	IN	50	IN	L
2	2021	MI	PT	Business Analyst	40000	USD	40000	AU	20	AU	M
3	2022	MI	FT	Product Manager	90000	USD	90000	DE	60	DE	L
4	2023	MI	PT	Part-Time Data Analyst	20000	USD	20000	US	50	US	M
5	2023	MI	PT	Part-Time Data Analyst	22000	USD	22000	US	50	US	M
6	2023	MI	PT	Part-Time Data Analyst	21000	USD	21000	CA	60	CA	L
7	2021	MI	FT	Data Analyst	NULL	NULL	60000	US	50	US	NULL
8	2022	MI	FT	Data Analyst	NULL	NULL	70000	US	60	US	NULL
9	2023	MI	FT	Data Analyst	NULL	NULL	80000	US	70	US	NULL

20. Guiding clients in switching domains based on salary insights:

- DECLARE

```
    @emp_experience_level VARCHAR(5) = 'MI',
    @emp_job_title VARCHAR(100) = 'Data Analyst',
    @emp_location VARCHAR(10) = 'US';
-- Step 1: Find employee's current average salary
SELECT
    job_title AS recommended_job_title,
    AVG(salary_in_usd) AS recommended_avg_salary
FROM workforce
WHERE employee_residence = @emp_location
  AND job_title <> @emp_job_title
GROUP BY job_title
HAVING
    AVG(salary_in_usd) >
    (
        SELECT AVG(salary_in_usd)
```

```
        FROM workforce
        WHERE job_title = @emp_job_title
          AND experience_level = @emp_experience_level
          AND employee_residence = @emp_location
    )
ORDER BY recommended_avg_salary DESC;
```

- Output:

	recommended_job_title	recommended_avg_salary
1	Engineering Manager	180000
2	Data Scientist	150000